

Block-level characterization

$$n_{\mathrm{ON},j}^{FC}(k), \alpha_j(k), \beta_j(k), Y_j^{FC}(k), W_{P,j}^{FC}(k)$$

Input Parameters

$P_{\max,j}$ $Type$

$W_{\min,j}$ $W_{\max,j}$

$Y_{\min,j}$ $Y_{\max,j}$

Day-Ahead Optimization

Decision Variables	Device-level constraint		Kitchen-level constraint
$E^{rlx}(k)$	$W_{T,j}(k) = W_{T,j}(k-1) + \lvert\mathcal{T}_j(k)\rvert\alpha_j(k) - n_{\mathrm{ON},j}^{OPT}(k)(\alpha_j(k)+\beta_j(k))$		$E^{NF}(k) + E^{TF}(k) + E^{PF}(k) \leq E^{lim}(k) + E^{rlx}(k)$
$Y_j^{OPT}(k)$	$W_{\min,j} \leq W_{T,j}(k) \leq W_{max,j}$	$Y_{\min,j} \leq Y_j^{OPT}(k) \leq Y_{\max,j}$	**Objective Function**
$n_{\mathrm{ON},j}^{OPT}(k)$			
$W_{T,j}^{REF}(k)$	$W_{P,j}(k) = W_{\min,j} + \frac{W_{\max,j} - W_{\min,j}}{Y_{\max,j} - Y_{\min,j}}\left(Y_j^{OPT}(k) - Y_{\min,j}\right)$		$\min\ C_{\mathrm{obj}} + \lambda_{\mathrm{rlx}} C_{\mathrm{violation}}$
$W_{P,j}^{REF}(k)$			